

A First Course in Causal Inference Ch 7

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Ch. 7 Matched-Pairs Experiment

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- ▶ The matched-pairs experiment (MPE) is the most extreme version of the SRE with only one treated unit and one control unit within each stratum.
- ▶ It has its own estimation and inference strategy

7.1 Design of the experiment and potential outcomes

- ▶ Consider an experiment with $2n$ units.
- ▶ If we have predictive covariates to the outcomes, we can pair units based on the similarity of covariates.
- ▶ With many covariates, we can define pairwise distances between units and the form pairs based on these distances.
- ▶ We assume that the pairs are formed based on the covariates and the discuss the subsequent design and analysis issues.

7.1 Design of the experiment and potential outcomes

Setup

- ▶ Let (i, j) index the unit j in pair i , where $i = 1, \dots, n$ and $j = 1, 2$.
- ▶ Unit (i, j) has potential outcomes $Y_{ij}(1)$ and $Y_{ij}(0)$ under the treatment and control, respectively.
- ▶ Within each pair, we randomly assign one into the treatment group ($Z_i = 1$) and the control group ($Z_i = 0$).

7.1 Design of the experiment and potential outcomes

Definition 7.1 (MPE)

The treatment assignments satisfy

$$(Z_i)_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}\left(\frac{1}{2}\right).$$

The observed outcomes within pair i are

$$Y_{i1} = Z_i Y_{i1}(1) + (1 - Z_i) Y_{i1}(0) = \begin{cases} Y_{i1}(1), & \text{if } Z_i = 1, \\ Y_{i1}(0), & \text{if } Z_i = 0, \end{cases}$$

and

$$Y_{i2} = Z_i Y_{i2}(0) + (1 - Z_i) Y_{i2}(1) = \begin{cases} Y_{i2}(0), & \text{if } Z_i = 1, \\ Y_{i2}(1), & \text{if } Z_i = 0. \end{cases}$$

So the observed data are $(Z_i, Y_{i1}, Y_{i2})_{i=1}^n$

7.3 Neymanian inference

- ▶ The average causal effect within pair i is

$$\tau_i = \frac{1}{2} [\{Y_{i1}(1) - Y_{i2}(0)\} + \{Y_{i2}(1) - Y_{i1}(0)\}].$$

- ▶ and the average causal effect for all units is

$$\tau = \frac{1}{n} \sum_{i=1}^n \tau_i = \frac{1}{2n} \sum_{i=1}^n \sum_{j=1}^2 \{Y_{ij}(1) - Y_{ij}(0)\}.$$

- ▶ Both $\hat{\tau}_i$ and $\hat{\tau}$ are unbiased for τ_i and τ , respectively.

7.3 Neymanian inference

We can calculate the variance of $\hat{\tau}$.

For simplicity, let $A_i = Y_{i1}(1) - Y_{i2}(0)$ and $B_i = Y_{i2}(1) - Y_{i1}(0)$. Then

$$\hat{\tau}_i = Z_i A_i + (1 - Z_i) B_i = Z_i (A_i - B_i) + B_i.$$

Check the unbiasedness of $\hat{\tau}_i$:

$$\mathbb{E}[\hat{\tau}_i] = \mathbb{E}[Z_i](A_i - B_i) + B_i = \frac{1}{2}(A_i - B_i) + B_i = \frac{A_i + B_i}{2}. \quad (\because Z_i \sim \text{Bernoulli}(1/2).)$$

$$\text{Here, } \text{var}(\hat{\tau}_i) = \text{var}(Z_i(A_i - B_i) + B_i) = \text{var}(Z_i)(A_i - B_i)^2 = \frac{1}{4}(A_i - B_i)^2.$$

Therefore,

$$\text{var}(\hat{\tau}) = \text{var}\left(\frac{1}{n} \sum_{i=1}^n \text{var}(\hat{\tau}_i)\right) = \frac{1}{4n^2} \sum_{i=1}^n \{Y_{i1}(1) - Y_{i2}(0) + Y_{i1}(0) - Y_{i2}(1)\}. \quad \square$$

7.3 Neymanian inference

- ▶ We cannot follow the same strategy under the SRE to estimate the variance of $\hat{\tau}$ under the MPE.
- ▶ In MPE, the within-pair sample variances of the outcomes are not well defined.
 - ▶ In SRE, we calculate variances for a treatment group and a control group in each stratum but in MPE, there is only one in both groups in each stratum.
- ▶ Is it possible to estimate the variance of $\hat{\tau}$ in the MPE?

7.3 Neymanian inference

- ▶ Think about the classic i.i.d. sampling:
- ▶ Suppose the $\hat{\tau}_i$'s are i.i.d. with mean μ and σ^2 .
- ▶ Then the variance of $\hat{\tau} = n^{-1} \sum_{i=1}^n \hat{\tau}_i$ is σ^2/n .
- ▶ An unbiased estimator for σ^2 is $(n-1)^{-1} \sum_{i=1}^n (\hat{\tau}_i - \hat{\tau})^2$.
- ▶ So the unbiased estimator for $\text{var}(\hat{\tau})$ is

$$\hat{V} = \frac{1}{n(n-1)} \sum_{i=1}^n (\hat{\tau}_i - \hat{\tau})^2. \quad (1)$$

- ▶ We derive $\hat{V}$ from a completely different setting from the MPE and a different motivation, so does it work for the MPE?

7.3 Neymanian inference

Theorem 7.1

Under the MPE, $\widehat{V}$ defined in (1) is a conservative estimator for the true variance of $\hat{\tau}$:

$$\mathbb{E}[\widehat{V}] - \text{var}(\hat{\tau}) = \frac{1}{n(n-1)} \sum_{i=1}^n (\tau_i - \tau)^2 \geq 0.$$

If the τ_i 's are constant across pairs, then $\mathbb{E}[\widehat{V}] = \text{var}(\hat{\tau})$.

► Thm. 7.1 states that under the MPE,

1. $\widehat{V}$ is a conservative variance estimator in general;
2. $\widehat{V}$ is unbiased if the average causal effects are constant across pairs.

7.3 Neymanian inference

Proof of Thm. 7.1:

Recall $\sum_{i=1}^n (a_i - \bar{a})^2 = \sum_{i=1}^n a_i^2 - n\bar{a}^2$ and $\text{var}[\mathcal{W}] = \mathbb{E}[\mathcal{W}^2] - (\mathbb{E}[\mathcal{W}])^2$. Use them in the following manipulation:

$$\begin{aligned} n(n-1)\mathbb{E}[\hat{V}] &= \mathbb{E}\left[\sum_{i=1}^n (\hat{\tau}_i - \hat{\tau})^2\right] = \mathbb{E}\left[\sum_{i=1}^n \hat{\tau}_i^2 - n\hat{\tau}^2\right] \\ &= \sum_{i=1}^n \{\text{var}(\hat{\tau}) + \tau_i^2\} - n\{\text{var}(\hat{\tau}) + \tau^2\} \\ &= n^2 \text{var}(\hat{\tau}) - n \text{var}(\hat{\tau}) + \sum_{i=1}^n (\tau_i - \tau)^2. \end{aligned}$$

Therefore,

$$\mathbb{E}[\hat{V}] = \text{var}(\hat{\tau}) + \{n(n-1)\}^{-1} \sum_{i=1}^n (\tau_i - \tau)^2 \geq 0. \quad \square$$

7.3 Neymanian inference

Both the point estimator $\hat{\tau}$ and the variance estimator $\hat{V}$ can be conveniently obtained by OLS:

Proposition 7.1

$\hat{\tau}$ and $\hat{V}$ are identical to the coefficient and variance estimator of the intercept from the OLS fit of the vector $(\hat{\tau}_1, \hat{\tau}_2, \dots, \hat{\tau}_n)^T$ on the intercept only.

7.3 Neymanian inference

Proof of Prop. 7.1.

7.4 Covariate adjustment

- ▶ Although we have matched on covariates in the design stage, it is possible that the matching is not perfect ($X_{i1} \neq X_{i2}$).
- ▶ In this case, we can adjust for the covariates to further improve estimation efficiency.
- ▶ Assume that each unit (i, j) has covariates X_{ij} .

7.4 Covariate adjustment

- ▶ Define the within-pair differences in covariates:

$$\hat{\tau}_{X,i} = Z_i(X_{i1} - X_{i2}) + (1 - Z_i)(X_{i2} - X_{i1}) = (2Z_i - 1)(X_{i1} - X_{i2}),$$

- ▶ and their average:

$$\hat{\tau}_X = \frac{1}{n} \sum_{i=1}^n \hat{\tau}_{X,i}.$$

- ▶ We can show that

$$\mathbb{E}[\hat{\tau}_{X,i}] = 0, \quad \mathbb{E}[\hat{\tau}_X] = 0,$$

- ▶ and

$$\text{cov}(\hat{\tau}_X) = \frac{1}{n^2} \sum_{i=1}^n \hat{\tau}_{X,i} \hat{\tau}_{X,i}^T.$$

7.4 Covariate adjustment

- ▶ In a realized MPE, $\text{cov}(\hat{\tau}_X)$ is not zero unless all the $\hat{\tau}_X$'s are zero.
 $\rightsquigarrow$ Some realized values of $\hat{\tau}_X \neq 0$.
- ▶ This is the reason why there exists imbalance and we would like to adjust the imbalance to improve estimation efficiency.
- ▶ Similar to the discussion in the last chapter, consider a class of estimators:

$$\hat{\tau}_\gamma = \hat{\tau} - \gamma^T \hat{\tau}_X,$$

which $\hat{\tau}_\gamma - \hat{\tau}$ has mean zero for any fixed γ .

- ▶ We want to choose γ to minimize the variance of $\hat{\tau}(\gamma)$.

7.4 Covariate adjustment

- ▶ The variance of $\hat{\tau}(\gamma)$ is

$$\begin{aligned}\text{var} &= \text{var}(\hat{\tau} - \gamma^T \hat{\tau}_X) \\ &= \text{var}(\hat{\tau}) + \gamma^T \text{cov}(\hat{\tau}_X) \gamma - 2\gamma^T \text{cov}(\hat{\tau}_X, \hat{\tau}),\end{aligned}$$

which is minimized at

$$\tilde{\gamma} = \text{cov}(\hat{\tau}_X)^{-1} \text{cov}(\hat{\tau}_X, \hat{\tau}).$$

- ▶ Rewrite the formula for $\text{cov}(\hat{\tau}_X)$ as

$$\text{cov}(\hat{\tau}_X) = \frac{1}{n^2} \sum_{i=1}^n |\hat{\tau}_{X,i}| |\hat{\tau}_{X,i}|^T,$$

where $|\cdot|$ denotes the component-wise absolute value of a vector.

7.4 Covariate adjustment

- ▶ We can calculate $\text{cov}(\hat{\tau}_X)$ but we cannot calculate $\text{cov}(\hat{\tau}_X, \hat{\tau})$
 - ▶ This is because we need unknown potential outcomes to compute this.
- ▶ Fortunately, we can obtain an unbiased estimator for it.

Theorem 7.2

An unbiased estimator for $\text{cov}(\hat{\tau}_X, \hat{\tau})$ is

$$\hat{\theta} = \frac{1}{n(n-1)} \sum_{i=1}^n (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau}).$$

7.4 Covariate adjustment

Substituting $\hat{\theta}$ into $\tilde{\gamma}$, then we obtain $\hat{\gamma}$ as follows:

$$\begin{aligned}\hat{\gamma} &= \left(\frac{1}{n^2} \sum_{i=1}^n \hat{\tau}_{X,i} \hat{\tau}_{X,i}^T \right)^{-1} \left\{ \frac{1}{n(n-1)} \sum_{i=1}^n (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau}) \right\} \\ &= \left(\frac{1}{n^2} \sum_{i=1}^n (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_{X,i} - \hat{\tau}_X)^T + \frac{1}{n^2} \cdot n \hat{\tau}_X \hat{\tau}_X^T \right)^{-1} \left\{ \frac{1}{n(n-1)} \sum_{i=1}^n (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau}) \right\} \\ &\approx \left(\sum_{i=1}^n (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_{X,i} - \hat{\tau}_X)^T \right)^{-1} \{ (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau}) \}.\end{aligned}$$

The second equality follows from $\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$. The approximation holds from $\frac{n^2}{n(n-1)} \xrightarrow{n \rightarrow \infty} 1$ and $\frac{n \hat{\tau}_X \hat{\tau}_X^T}{n^2} \xrightarrow{n \rightarrow \infty} 0$.

7.4 Covariate adjustment

- ▶ This is approximately the coefficient of the $\hat{\tau}_{X,i}$ in the OLS fit of the $\hat{\tau}_i$'s on the $\hat{\tau}_{X,i}$'s with an intercept.
- ▶ The final estimator is

$$\hat{\tau}_{\text{adj}} = \hat{\tau} - \hat{\gamma}^T \hat{\tau}_X,$$

which is approximately the intercept in the OLS fit of the $\hat{\tau}_i$'s on the $\hat{\tau}_{X,i}$'s with an intercept.

- ▶ The conservative variance estimator for $\hat{\tau}_{\text{adj}}$ is then

$$\begin{aligned}\hat{V}_{\text{adj}} &= \text{var}(\hat{\tau} - \hat{\gamma}^T \hat{\tau}_X) \\ &= \hat{V} + \hat{\gamma}^T \text{cov}(\hat{\tau}_X) \hat{\gamma} - 2\hat{\gamma}^T \hat{\theta} \\ &= \hat{V} - \hat{\theta} \text{cov}(\hat{\tau}_X)^{-1} \hat{\theta}.\end{aligned}$$

7.4 Covariate adjustment

Proposition 7.2

Under the MPE, the covariance-adjusted estimator $\hat{\tau}_{\text{adj}}$ and the associated variance estimator $\hat{V}_{\text{adj}}$ can be conveniently approximated by the intercept and the associated variance estimator from the OLS fit of the vector of the $\hat{\tau}_i$'s on the 1's and the matrix of the $\hat{\tau}_{X,i}$'s.

- ▶ Neither Prop. 7.1 nor 7.2 requires the EHW correction of the variance estimator.
- ▶ In the MPE, we only use a pair so we do not need to center the covariate.
- ▶ Explicitly, let $\bar{X}$ be the mean covariate, then
$$X_{i1} - X_{i2} = (X_{i1} - \bar{X}) - (X_{i2} - \bar{X}).$$

7.6 Comparing the MPE and CRE.

